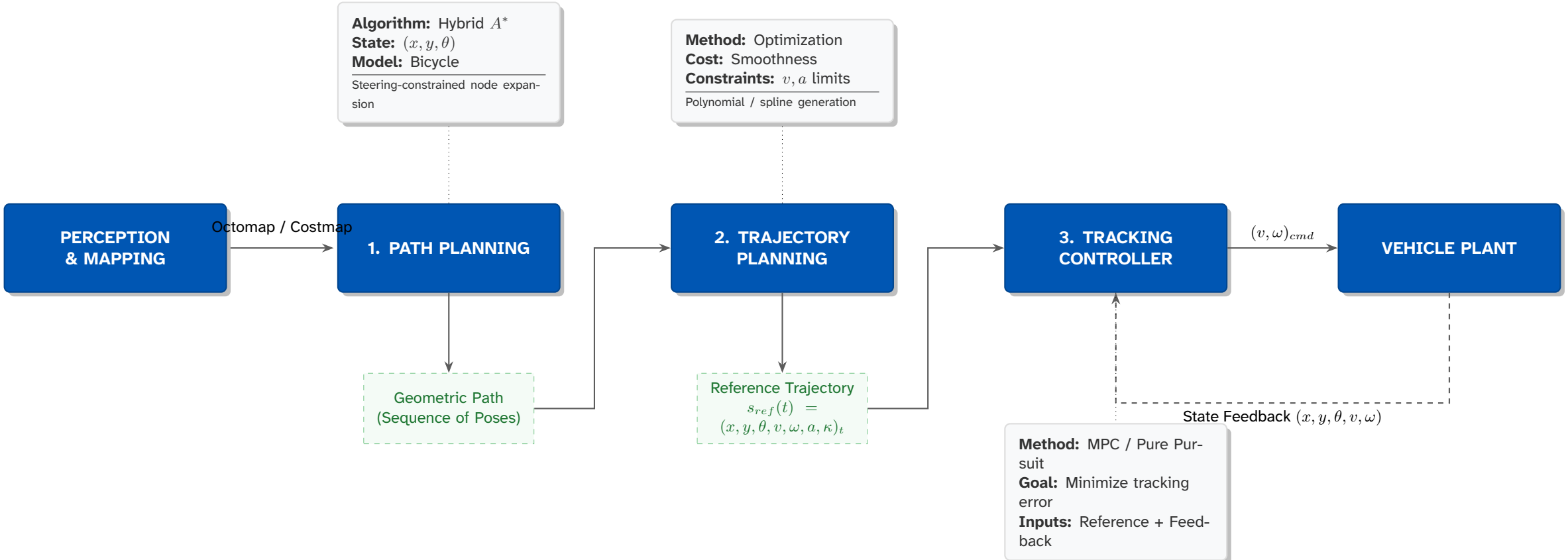
Algorithm: Hybrid $A^*$
State: $(x, y, \theta)$
Model: Bicycle
Steering-constrained node expansion
Method: Optimization
Cost: Smoothness
Constraints: $v, a$ limits
Polynomial / spline generation
PERCEPTION & MAPPING
Octomap / Costmap
1. PATH PLANNING
2. TRAJECTORY PLANNING
3. TRACKING CONTROLLER
$(v, \omega)_{cmd}$
VEHICLE PLANT
Geometric Path (Sequence of Poses)
Reference Trajectory $s_{ref}(t) = (x, y, \theta, v, \omega, a, \kappa)_t$
State Feedback $(x, y, \theta, v, \omega)$
Method: MPC / Pure Pursuit
Goal: Minimize tracking error
Inputs: Reference + Feedback